1. INTRODUCTION

Marine plastic pollution has emerged as one of the most significant environmental challenges affecting oceans and aquatic ecosystems worldwide. Large quantities of plastic waste enter water bodies every year, posing serious threats to marine life, biodiversity, and environmental sustainability. Plastics are non-biodegradable and can persist in the environment for decades, causing harm to aquatic organisms through ingestion, entanglement, and habitat degradation. Monitoring and removing plastic waste from marine environments is therefore essential for preserving ecosystem health and maintaining ecological balance.

Traditional methods of detecting plastic waste in underwater environments rely heavily on manual inspection, which is time-consuming, labor-intensive, and often inefficient. The advancement of Artificial Intelligence (AI) and Computer Vision technologies has enabled the development of automated systems capable of identifying and locating plastic waste with high accuracy. Object detection models can analyze images and detect specific objects within them, making them suitable for environmental monitoring applications.

The proposed system, Plastic Detection Using YOLO, utilizes the YOLOv8 (You Only Look Once) object detection algorithm to automatically detect plastic waste in underwater images. The system is trained using the TrashCan dataset, which contains annotated images of marine debris. The trained model is integrated with a Streamlit-based web application that allows users to upload underwater images and obtain detection results instantly. By combining deep learning techniques with a user-friendly interface, the system provides an efficient solution for identifying plastic waste in marine environments and supporting environmental conservation efforts.

1.1 PROBLEM STATEMENT

Marine plastic pollution has become a major environmental concern due to the increasing accumulation of plastic waste in oceans, rivers, and other aquatic ecosystems. Existing methods of detecting underwater plastic waste primarily depend on manual observation and analysis, which are often time-consuming, expensive, and prone to human error. The need for an automated, accurate, and efficient system for identifying plastic waste in underwater images has become increasingly important.

The objective of this project is to develop an intelligent plastic detection system using the YOLOv8 object detection algorithm that can automatically identify and localize plastic waste in underwater images. The system aims to improve detection efficiency, reduce human intervention, and support environmental monitoring and marine conservation activities.

1.2 OBJECTIVE

The main objectives of the Plastic Detection Using YOLO system are:

- To develop an automated system for detecting plastic waste in underwater images.
- To train a YOLOv8-based object detection model using a marine debris dataset.
- To accurately identify and localize plastic objects using bounding boxes and confidence scores.
- To reduce the dependency on manual monitoring and inspection methods.
- To provide a user-friendly Streamlit interface for image upload and detection.
- To improve the efficiency and reliability of marine plastic waste monitoring.
- To contribute towards environmental conservation through the application of Artificial Intelligence and Computer Vision technologies.

1.3 SCOPE

The scope of the Plastic Detection Using YOLO project focuses on the automated detection of plastic waste present in underwater images. The system utilizes deep learning-based object detection techniques to identify and locate plastic objects with minimal human intervention. The project covers dataset preparation, annotation conversion, model training, testing, and deployment through a web-based interface.

The system can be used by environmental researchers, marine conservation organizations, and educational institutions for monitoring marine pollution. It provides a foundation for future enhancements such as real-time video analysis, multi-class waste classification, integration with underwater robotic systems, and large-scale marine monitoring applications. Although the current implementation focuses on plastic detection from static underwater images, the proposed framework can be extended to support broader environmental monitoring tasks in the future.

1.4 ORGANISATION OF REPORT

This report is organized into seven major chapters that describe the design, development, implementation, and evaluation of the Plastic Detection Using YOLO system.

Chapter 1 – Introduction presents the background of marine plastic pollution, problem statement, objectives, scope, and an overview of the proposed system.

Chapter 2 – Literature Survey discusses related research works, existing waste detection systems, deep learning techniques, and technologies used in the project.

Chapter 3 – System Analysis describes the existing system, its limitations, the proposed YOLO-based plastic detection system, its advantages, applications, and software requirements.

Chapter 4 – System Design presents the UML diagrams and Data Flow Diagrams (DFDs) that illustrate the system architecture, workflow, and data movement.

Chapter 5 – Implementation explains the implementation of dataset preprocessing, annotation conversion, YOLOv8 model training, plastic detection, and Streamlit-based deployment.

Chapter 6 – Testing presents the test cases, validation procedures, performance evaluation, and testing outcomes used to verify system functionality and reliability.

Chapter 7 – Results discusses the detection results, performance analysis, application screenshots, and interpretation of the obtained outputs.

Finally, the report concludes with the project outcomes, future enhancement possibilities, references, appendix, and project mapping resources associated with the Plastic Detection Using YOLO system.

2. LITERATURE SURVEY

2.1 RESEARCH PAPERS AND PUBLICATIONS

Table 2.1: Research papers and publications related to Plastic Detection using YOLO

S.No	Title	Authors	Year	Outcomes
1	TrashCan: A Semantically-Segmented Dataset towards Visual Detection of Marine Debris	Jungseok Hong, Michael Fulton, Junaed Sattar	2020	Introduced the TrashCan dataset containing annotated underwater images of marine debris. It serves as a benchmark dataset for marine debris detection and segmentation research.
2	State of the Art Applications of Deep Learning within Tracking and Detecting Marine Debris: A Survey	Zoe Moorton, Zeyneb Kurt, Wai Lok Woo	2024	Surveyed recent deep learning approaches for marine debris detection and found that YOLO-based methods provide strong performance in terms of accuracy and speed.
3	A Dataset for Detection and Segmentation of Underwater Marine Debris in Shallow Waters	SeaClear Research Team	2024	Developed a large-scale annotated underwater debris dataset to improve detection and segmentation models for marine waste monitoring.
4	Marine Debris Detection Model	J. Liu, Y. Zhou	2023	Proposed an enhanced YOLOv5 model that

	Based on the Improved YOLOv5			improved marine debris detection accuracy and robustness in underwater environments.
5	An Improved YOLOv5-Based Underwater Object Detection Framework	Jian Zhang, Jinshuai Zhang, Kexin Zhou, Yonghui Zhang, Hongda Chen, Xinyue Yan	2023	Improved underwater object detection performance through better feature extraction and localization mechanisms.
6	YOLOv8-C2f-Faster-EMA: An Improved Underwater Trash Detection Model Based on YOLOv8	Jin Zhu, Tao Hu, Linhan Zheng, Nan Zhou, Huilin Ge, Zhichao Hong	2024	Enhanced YOLOv8 architecture for underwater trash detection, achieving higher accuracy and faster inference compared to standard YOLOv8.
7	Research on the Identification and Classification of Marine Debris Based on Improved YOLOv8	Wenbo Jiang, Lusong Yang, Yun Bu	2024	Improved marine debris classification and detection using modifications to YOLOv8, resulting in better generalization and recognition performance.

Marine plastic pollution has become a significant environmental concern, leading researchers to develop automated systems for detecting marine debris. Early research focused on creating benchmark datasets such as the TrashCan dataset, which enabled the training and evaluation of deep learning models for underwater waste detection. Recent studies have increasingly adopted YOLO-based object detection techniques due to their ability to perform real-time detection with high accuracy.

Several researchers have proposed improvements to YOLOv5 and YOLOv8 architectures to address challenges such as poor underwater visibility, small object detection, and complex backgrounds. These studies demonstrate that deep learning-based object detection models can significantly improve the efficiency of marine debris monitoring. Motivated by these findings, the proposed project employs YOLOv8 and the TrashCan dataset to accurately detect and localize plastic waste in underwater images through a user-friendly Streamlit application.

2.2 TECHNOLOGY

Plastic Detection Using YOLO is developed using modern Artificial Intelligence, Computer Vision, and Web Application technologies that provide an efficient, accurate, and user-friendly platform for marine plastic waste detection. The system follows a machine learning-based architecture, where the trained YOLO model performs object detection, while the Streamlit interface enables users to interact with the system through image uploads and result visualization.

1. YOLOv8: YOLOv8 (You Only Look Once Version 8) is the core object detection model used in the project. It performs real-time object detection by identifying plastic waste in underwater images and generating bounding boxes with confidence scores. YOLOv8 offers high detection accuracy and fast inference speed, making it suitable for environmental monitoring applications.

2. Ultralytics Framework: The Ultralytics framework is used for training, validating, and deploying the YOLOv8 model. It provides an easy-to-use interface for dataset management, model training, prediction generation, and performance evaluation.

3. Streamlit: Streamlit is used to develop the web-based user interface of the system. It enables users to upload underwater images, execute the trained YOLO model, and visualize plastic detection results through a simple and interactive interface.

4. KaggleHub: KaggleHub is used for downloading and managing the TrashCan dataset directly within the development environment. It simplifies dataset acquisition and integration into the training pipeline.

5. Google Colab: Google Colab is used as the primary training environment for model development. It provides cloud-based GPU resources that significantly reduce model training time and support efficient experimentation with deep learning models.

6. Visual Studio Code (VS Code): Visual Studio Code is used for application development, model integration, testing, and deployment. It provides a flexible development environment with support for Python programming and Streamlit application execution.

The integration of these technologies enables the Plastic Detection Using YOLO system to provide accurate underwater plastic waste detection, efficient model deployment, and a user-friendly platform for environmental monitoring and marine conservation applications.

3. SYSTEM ANALYSIS

System Analysis is an important phase in software development that involves studying the existing system, identifying its limitations, and proposing an improved solution to address current challenges. In the context of marine pollution monitoring, efficient detection of plastic waste is essential for environmental conservation. This chapter discusses the existing methods for plastic waste detection, their limitations, the proposed YOLO-based detection system, its advantages, applications, and software requirements specifications.

3.1 Existing System

Traditional plastic waste detection methods primarily rely on manual observation, image inspection, and conventional image processing techniques. Environmental researchers and marine monitoring organizations often use human inspection to identify plastic debris in underwater environments. Some existing computer vision systems employ basic image classification algorithms that can determine whether plastic is present in an image but fail to provide accurate localization.

Conventional approaches face difficulties in handling underwater conditions such as low visibility, varying illumination, water turbidity, and complex backgrounds consisting of rocks, sand, vegetation, and marine organisms. As a result, detection accuracy is often reduced, making these systems less reliable for large-scale monitoring applications.

3.1.1 Limitations of Existing System

- Manual inspection is time-consuming and labor-intensive.
- Human observation is prone to errors and inconsistencies.
- Traditional image processing methods struggle in complex underwater environments.
- Many systems only classify images without locating plastic waste.
- Detection accuracy decreases under poor lighting and visibility conditions.
- Existing methods are difficult to scale for large marine monitoring operations.
- Real-time detection capabilities are limited or unavailable.
- High operational costs are associated with manual monitoring processes.

3.2 Proposed System

The proposed system, Plastic Detection Using YOLO, utilizes a deep learning-based object detection approach to automatically identify and localize plastic waste in underwater images. The system employs the YOLOv8 object detection model trained on the TrashCan dataset containing annotated images of marine debris.

Users can upload underwater images through a Streamlit-based web interface. The uploaded image is processed by the trained YOLO model, which detects plastic waste and generates bounding boxes along with confidence scores. The detection results are then displayed through the user interface, providing an efficient and automated solution for marine plastic monitoring.

The proposed system combines Computer Vision, Deep Learning, and Web Technologies to improve detection accuracy and reduce the dependency on manual monitoring methods.

3.2.1 Advantages of Proposed System

- Fully automated plastic waste detection process.
- Accurate localization using bounding boxes and confidence scores.
- Faster detection compared to manual inspection methods.
- Improved performance in complex underwater environments.
- User-friendly Streamlit interface for easy interaction.
- Supports real-time image-based detection.
- Reduces human effort and operational costs.
- Scalable solution for large-scale marine monitoring applications.
- Can be integrated with underwater robots and autonomous cleanup systems.
- Provides reliable environmental monitoring support.

3.3 Applications

The Plastic Detection Using YOLO system can be applied in various environmental and marine monitoring domains. Some major applications include:

- Marine pollution monitoring and assessment.
- Ocean cleanup and conservation projects.
- Underwater robotic inspection systems.
- Autonomous marine waste collection systems.
- Environmental research and academic studies.
- Coastal and offshore waste monitoring.
- Smart surveillance systems for aquatic ecosystems.
- Government and environmental agency monitoring programs.
- Marine biodiversity protection initiatives.
- Detection and analysis of underwater debris accumulation.

3.4 Software Requirements Specification

The Software Requirements Specification (SRS) defines the software components, development tools, and execution environment required for the successful implementation of the Plastic Detection Using YOLO system.

Software Requirements:

Table 3.1: Software Requirements

Component	Specification
Operating System	Windows 10/11, Linux, macOS
Programming Language	Python 3.x
Deep Learning Framework	Ultralytics YOLOv8
Development Environment	Google Colab, VS Code
Web Framework	Streamlit
Computer Vision Library	OpenCV
Visualization Library	Matplotlib
Dataset	TrashCan Dataset
Dataset Management	KaggleHub
Version Control	Git
Browser Support	Google Chrome, Microsoft Edge, Mozilla Firefox

Hardware Requirements:

Table 3.2: Hardware Requirements

Component	Specification
Processor	Intel Core i5 or above
RAM	Minimum 8 GB
Storage	Minimum 10 GB Free Space
GPU (Optional)	NVIDIA GPU for Faster Training
Internet Connection	Required for Dataset Download and Deployment

These requirements ensure smooth development, training, testing, and deployment of the Plastic Detection Using YOLO system.

4. SYSTEM DESIGN

System Design is the process of defining the architecture, components, modules, interfaces, and data flow of a software system. It provides a blueprint for the implementation of the Plastic Detection Using YOLO system by illustrating how different components interact with each other to perform plastic waste detection. The system design focuses on user interaction, image processing, object detection, and result visualization. Various UML diagrams and Data Flow Diagrams (DFDs) are used to represent the structure, behavior, and data movement within the system.

4.1 UML Diagrams

Unified Modeling Language (UML) diagrams are used to visualize the design and functionality of the Plastic Detection Using YOLO system. UML diagrams provide a graphical representation of the system's structure, workflow, and interactions between different components. They help in understanding the behavior of the system and serve as a foundation for implementation.

The UML diagrams used in this project are:

- UML Activity Diagram
- UML Use Case Diagram
- UML Class Diagram
- UML Sequence Diagram

These diagrams illustrate the workflow of plastic detection, user interactions, relationships among system components, and the sequence of operations performed during object detection.

4.1.1 Activity Diagram

The Activity Diagram represents the workflow of the system from image upload to result generation. It illustrates the sequence of activities involved in detecting plastic waste using the trained YOLOv8 model.

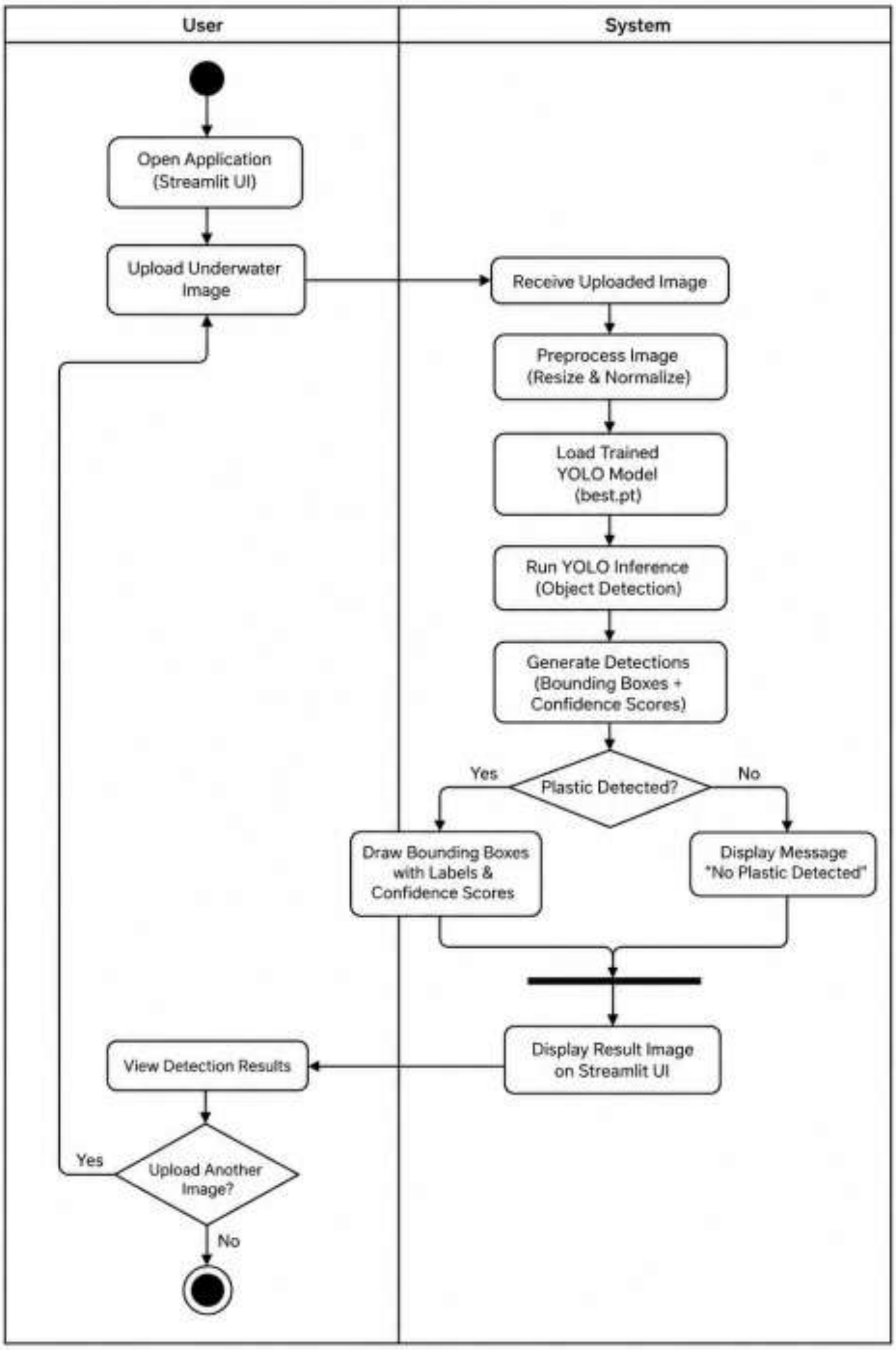


Figure 4.1: Activity Diagram

4.1.2 Use Case Diagram

The Use Case Diagram describes the interaction between the user and the system. It identifies the functionalities provided by the application and how users interact with them.

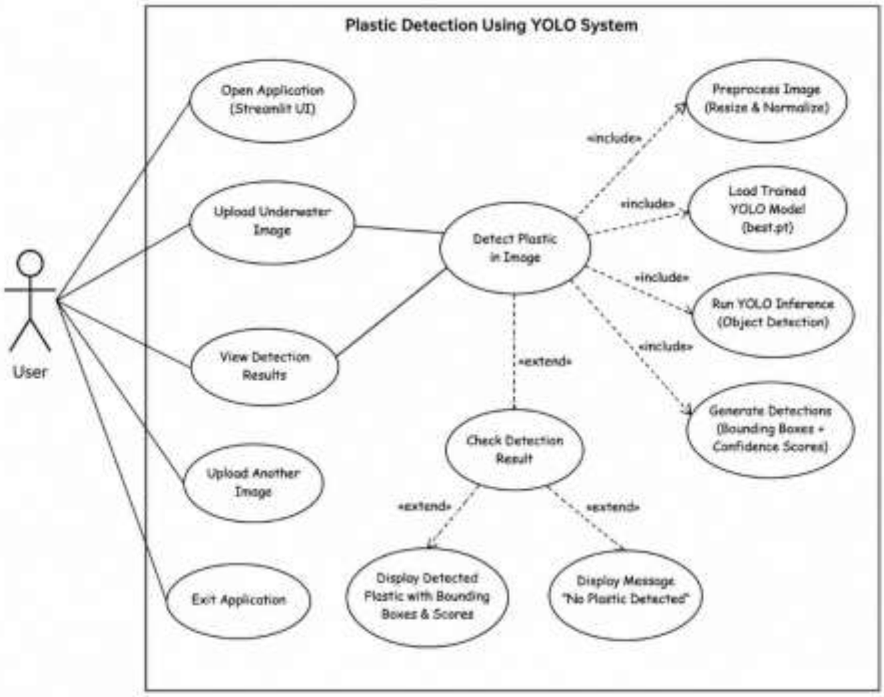


Figure 4.2: Use Case Diagram

4.1.3 Class Diagram

The Class Diagram represents the structural design of the system. It shows the major classes, their attributes, methods, and relationships involved in the plastic detection process.

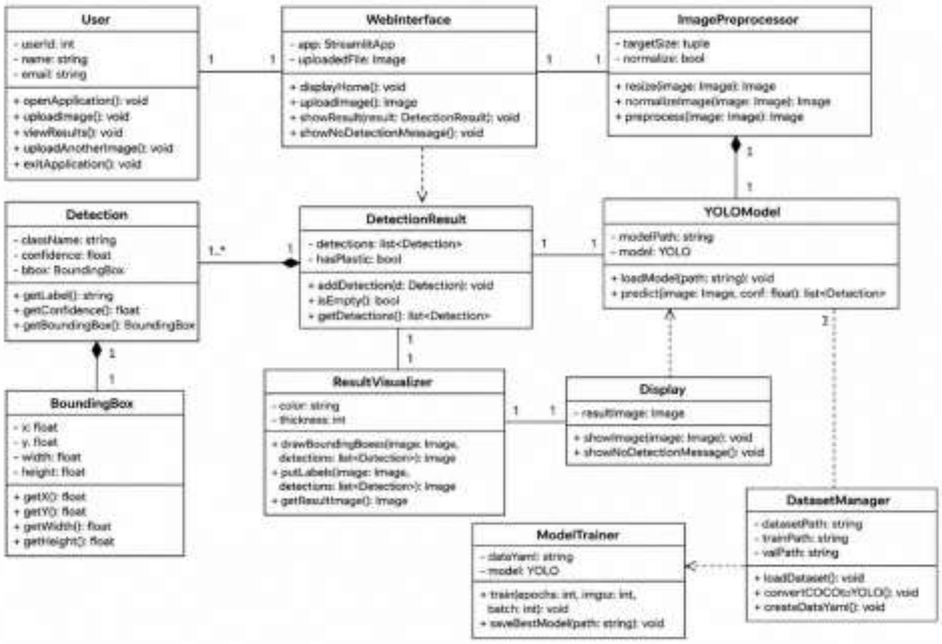


Figure 4.3: Class Diagram

4.1.4 Sequence Diagram

The Sequence Diagram illustrates the chronological flow of interactions between the user, Streamlit interface, YOLO model, and output components during the detection process.

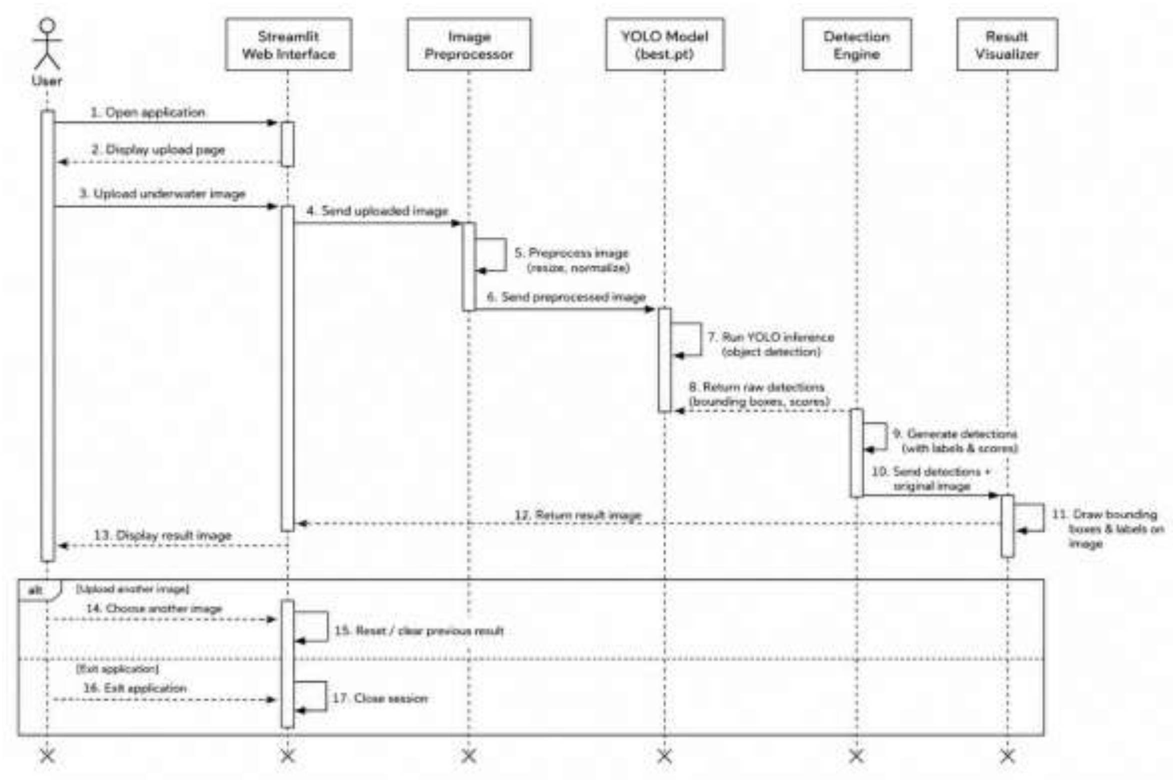


Figure 4.4: Sequence Diagram

4.2 Data Flow Diagrams

Data Flow Diagrams (DFDs) are used to represent the movement of data within the Plastic Detection Using YOLO system. They show how data enters the system, how it is processed, and how results are generated and returned to the user. DFDs help in understanding the functional flow and internal processing of the application.

The Data Flow Diagrams used in this project are:

- DFD Level 0
- DFD Level 1

4.2.1 DFD Level 0

The DFD Level 0 provides a high-level overview of the entire system. It represents the Plastic Detection Using YOLO system as a single process and illustrates the interaction between the user and the system.

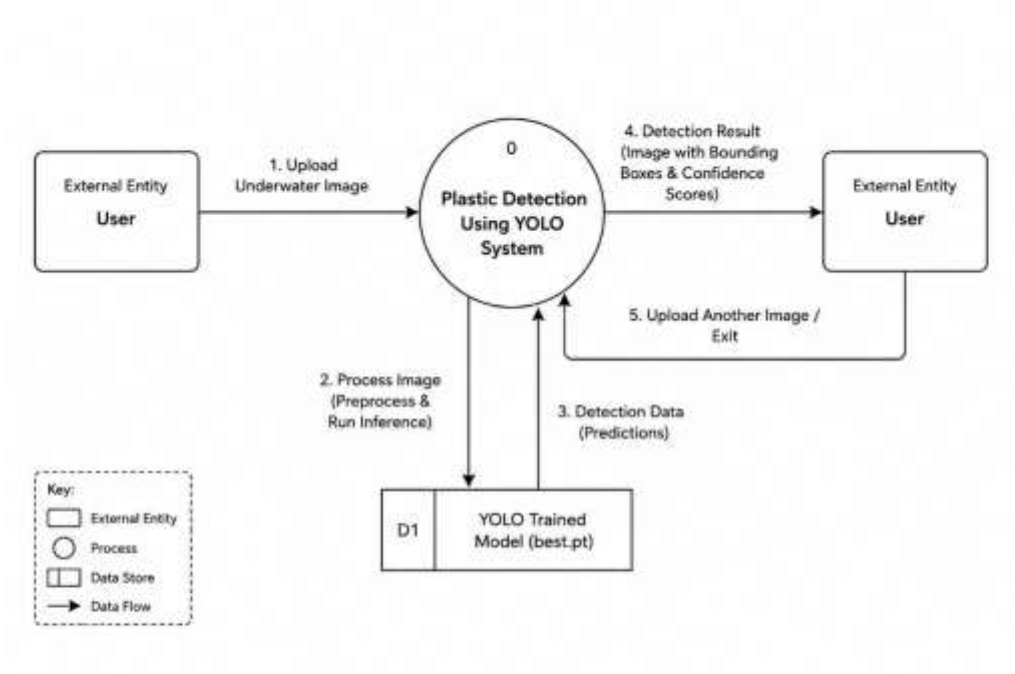


Figure 4.5: DFD Level 0 Diagram

4.2.2 DFD Level 1

The DFD Level 1 provides a detailed representation of the internal processes involved in image upload, preprocessing, YOLO-based detection, result generation, and visualization.

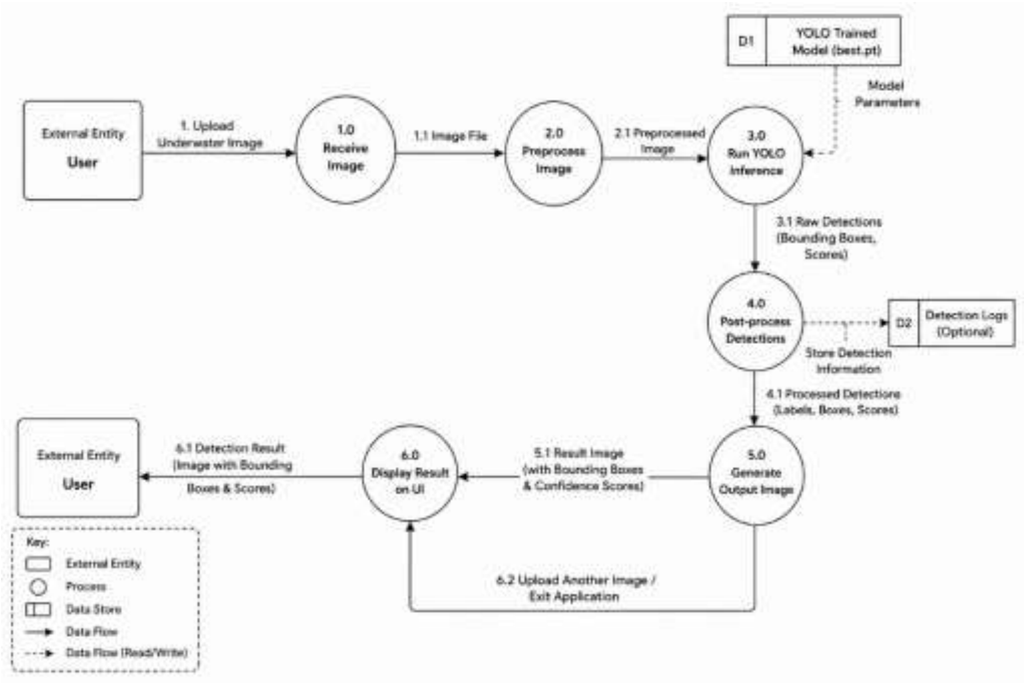


Figure 4.6: DFD Level 1 Diagram

The System Design chapter provides a comprehensive overview of the architecture and workflow of the Plastic Detection Using YOLO system. UML diagrams describe the system structure and behavior, while Data Flow Diagrams illustrate the movement and processing of data. Together, these diagrams provide a clear understanding of the design and operation of the proposed system.

5. IMPLEMENTATION

The implementation phase focuses on developing and integrating the various modules of the Plastic Detection Using YOLO system. The system combines deep learning, computer vision, and web technologies to provide an efficient solution for detecting plastic waste in underwater images. The implementation process includes dataset acquisition, data preprocessing, annotation conversion, model training, model evaluation, and deployment through a Streamlit-based web application.

5.1 Dataset Collection

The first stage of implementation involves collecting the dataset required for training the object detection model. The TrashCan dataset was obtained using KaggleHub and consists of underwater images containing marine debris and plastic waste. The dataset includes image annotations in COCO format, which provide information about the location of objects within each image.

The dataset serves as the foundation for training the YOLOv8 model to recognize and localize plastic waste in underwater environments.

5.2 Data Preprocessing

The collected dataset underwent preprocessing before being used for model training. Dataset exploration was performed to verify image quality, annotation consistency, and dataset structure.

The preprocessing activities included:

- Image inspection and visualization.
- Annotation verification.
- Organization of training and validation data.
- Preparation of dataset folders required by YOLOv8.

These steps ensured that the dataset was suitable for efficient model training and evaluation.

5.3 COCO to YOLO Annotation Conversion

The TrashCan dataset originally contained annotations in COCO format. Since YOLOv8 requires annotations in YOLO format, a conversion process was implemented.

The conversion module performs the following operations:

- Reads COCO annotation files.
- Extracts bounding box coordinates.
- Converts coordinates into YOLO format.
- Generates corresponding label files for each image.
- Organizes images and labels into training and validation directories.

This process prepares the dataset for compatibility with the YOLOv8 training framework.

5.4 YOLOv8 Model Training

The core component of the system is the YOLOv8 object detection model. The model was trained using the prepared dataset to learn the characteristics of underwater plastic waste.

Training was performed using the YOLOv8 Medium (YOLOv8m) architecture with the following configuration:

Table 5.1: YOLOv8 Model Configuration

Parameter	Value
Model	YOLOv8m
Epochs	20
Image Size	640 × 640
Batch Size	16
Patience	3
Device	GPU
Object Class	Plastic

Various data augmentation techniques such as scaling, translation, flipping, mosaic augmentation, and color transformations were applied to improve model generalization and detection performance.

5.5 Model Validation and Testing

After training, the model was evaluated using validation images from the dataset. The validation process was used to measure the model's ability to detect plastic waste in unseen images.

The trained model generated:

- Bounding boxes around detected plastic objects.
- Confidence scores for each detection.
- Visualized detection outputs for performance analysis.

Testing confirmed that the model was capable of identifying plastic waste under varying underwater conditions.

5.6 Model Deployment

After successful training and evaluation, the best-performing model weights were saved as best.pt. These weights were then integrated into a web-based application developed using Streamlit.

The deployment process involved:

- Loading the trained YOLOv8 model.
- Accepting image uploads from users.
- Running object detection on uploaded images.
- Displaying detection results with bounding boxes.

The deployment module enables users to interact with the trained model without requiring technical knowledge of machine learning.

5.7 Streamlit User Interface

A user-friendly web interface was developed using Streamlit to simplify interaction with the detection system.

The interface provides the following functionalities:

- Upload underwater images.
- Display uploaded images.
- Execute YOLOv8-based plastic detection.
- Display detection results with bounding boxes.
- Provide visual feedback for identified plastic waste.

The simple design allows users to perform plastic detection efficiently through a web browser.

5.8 Working of the System

The operational workflow of the Plastic Detection Using YOLO system is as follows:

- The user accesses the Streamlit application.
- An underwater image is uploaded through the interface.
- The image is passed to the trained YOLOv8 model.
- The model processes the image and identifies plastic waste.
- Bounding boxes and confidence scores are generated.
- Detection results are visualized and displayed to the user.
- The user analyzes the output for environmental monitoring purposes.

The implementation phase successfully integrates dataset processing, annotation conversion, YOLOv8 model training, validation, and deployment into a unified plastic detection system. The combination of deep learning and web technologies enables efficient identification of marine plastic waste and provides an accessible platform for environmental monitoring and conservation efforts.

6. TESTING

Testing is an essential phase in software development that ensures the developed system functions correctly, meets user requirements, and produces reliable results. The Plastic Detection Using YOLO system was tested to verify its functionality, accuracy, performance, and usability. Various test cases were conducted to validate image upload functionality, object detection performance, result generation, and system responsiveness.

6.1 Testing Objectives

The objectives of testing the Plastic Detection Using YOLO system are:

- To verify the correct functioning of all system modules.
- To validate image upload and processing functionality.
- To evaluate the accuracy of plastic waste detection.
- To ensure proper generation of bounding boxes and confidence scores.
- To identify and eliminate potential errors and bugs.
- To assess system reliability and usability.

6.2 Types of Testing Performed

6.2.1 Functional Testing

Functional testing was performed to verify that each module of the system operates according to the specified requirements. The testing focused on image upload, object detection, result generation, and output display functionalities.

6.2.2 Integration Testing

Integration testing was conducted to ensure smooth interaction between different components, including the Streamlit interface, YOLOv8 model, image processing module, and output visualization module.

6.2.3 Performance Testing

Performance testing evaluated the system's ability to process images efficiently and generate detection results within an acceptable time frame.

6.2.4 Validation Testing

Validation testing was performed using unseen underwater images to verify the model's capability to correctly identify plastic waste and generalize to new data.

6.3 Test Cases

Table 6.1: Test Cases

Test Case ID	Test Description	Input	Expected Result	Actual Result	Status
TC-01	Upload Valid Image	Underwater image containing plastic waste	Image uploaded successfully	Successful upload	Pass
TC-02	Upload Invalid File Format	PDF/Text file	System rejects unsupported file	Unsupported file rejected	Pass
TC-03	Plastic Detection	Image containing plastic waste	Plastic detected with bounding box	Plastic detected successfully	Pass
TC-04	No Plastic Detection	Image without plastic waste	No detection or appropriate output	Correct output generated	Pass
TC-05	Result Visualization	Detection output image	Bounding boxes displayed correctly	Correct visualization obtained	Pass
TC-06	Multiple Image Testing	Different underwater images	Consistent detection performance	Successful predictions generated	Pass

TC-07	Application Response	Image upload and prediction	Quick response and output generation	Acceptable response time	Pass

6.4 Testing Results

The testing process demonstrated that the Plastic Detection Using YOLO system successfully performs plastic waste detection in underwater images. The Streamlit interface correctly accepts image uploads and displays the corresponding detection outputs. The YOLOv8 model accurately identifies plastic objects and generates bounding boxes around detected regions.

The integration between the trained model and the web interface operated without significant issues. Testing with various underwater images confirmed the system's ability to handle different environmental conditions and image characteristics.

6.5 System Validation

System validation was conducted using validation images from the TrashCan dataset as well as additional test images. The trained YOLOv8 model successfully identified plastic waste objects and localized them accurately within the images.

The validation process confirmed:

- Correct image processing workflow.
- Successful object detection functionality.
- Accurate bounding box generation.
- Effective integration between all system components.
- Reliable performance across multiple test samples.

. The testing phase verified the functionality, reliability, and performance of the Plastic Detection Using YOLO system. All major modules operated successfully and produced the expected results.

7. RESULTS

The Results chapter presents the outcomes obtained from the implementation of the Plastic Detection Using YOLO system. The trained YOLOv8 model was evaluated using underwater images containing plastic waste, and the detection results were analyzed to assess the effectiveness of the proposed approach. The system successfully identified and localized plastic waste objects by generating bounding boxes and confidence scores through a user-friendly Streamlit interface.

7.1 Present Results Obtained

The Plastic Detection Using YOLO system was successfully developed, trained, and deployed. The YOLOv8 model demonstrated the ability to accurately detect plastic waste present in underwater images. During testing, the model generated bounding boxes around detected plastic objects and provided confidence scores indicating the certainty of each prediction.

The integration of the trained model with the Streamlit web application enabled users to upload underwater images and obtain detection results in real time. The system performed effectively across various test images and demonstrated reliable object detection capabilities under different underwater conditions.

The major outcomes achieved are:

- Successful training of the YOLOv8 model using the TrashCan dataset.
- Accurate detection and localization of plastic waste in underwater images.
- Generation of bounding boxes and confidence scores for detected objects.
- Development of a user-friendly Streamlit-based web application.
- Real-time prediction and result visualization.
- Effective integration of computer vision and deep learning techniques for environmental monitoring.

7.2 Screenshots

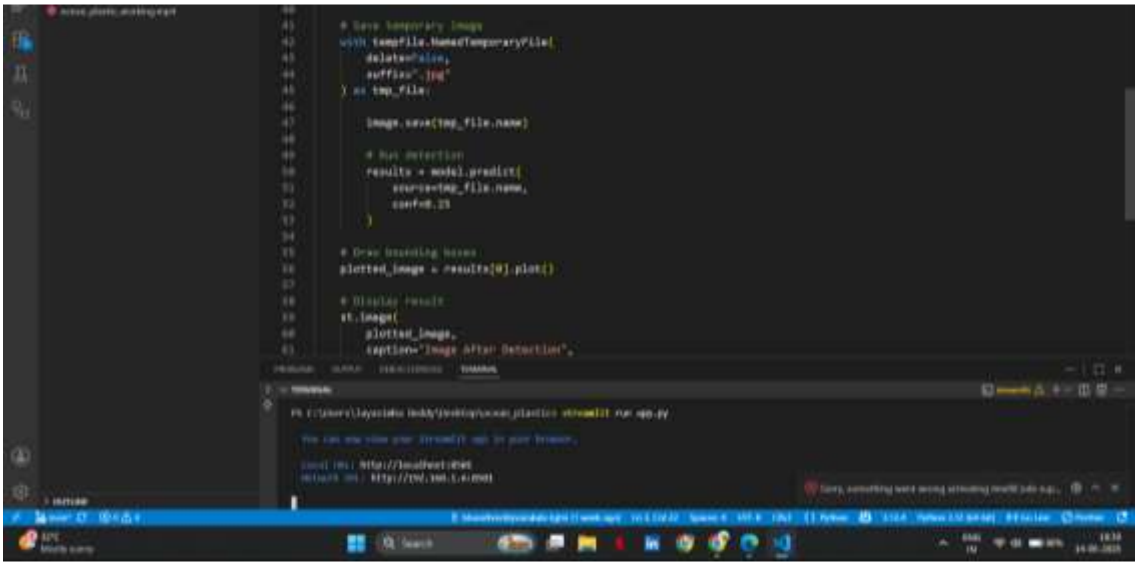


Figure 7.1: Successful compilation of code

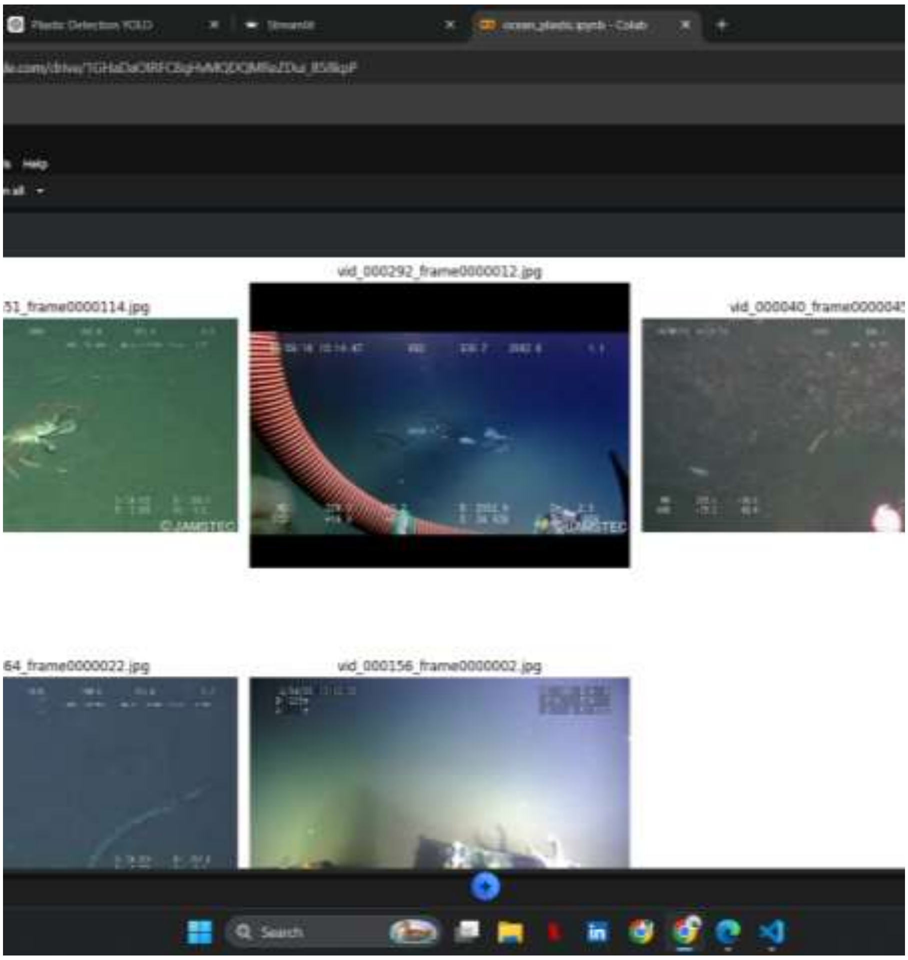


Figure 7.2: Testing of randomly selected images from validation set

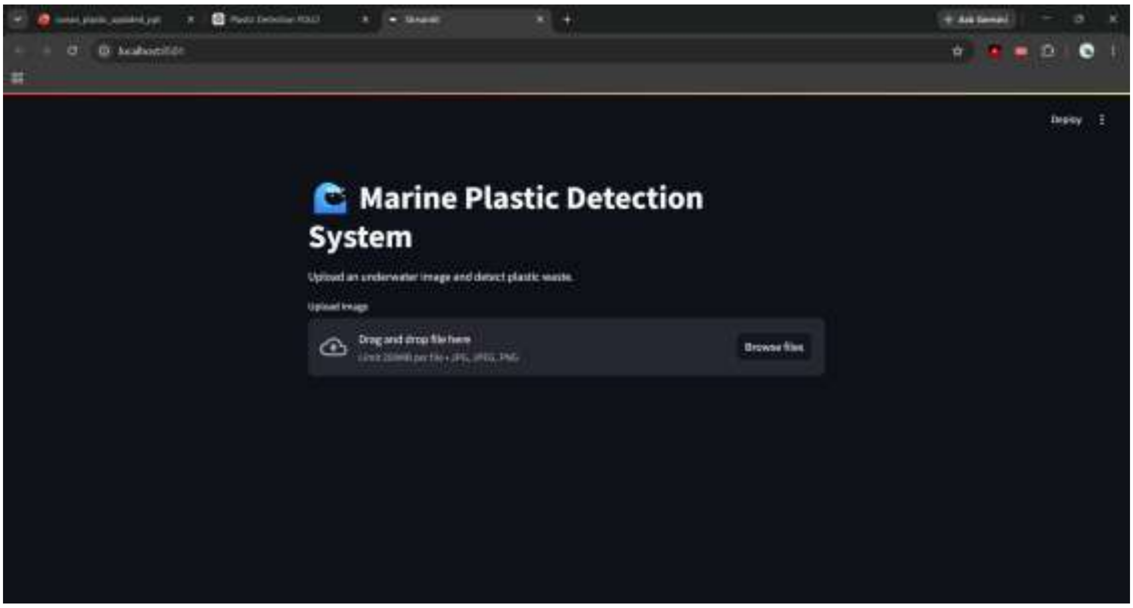


Figure 7.3: Streamlit web application landing page

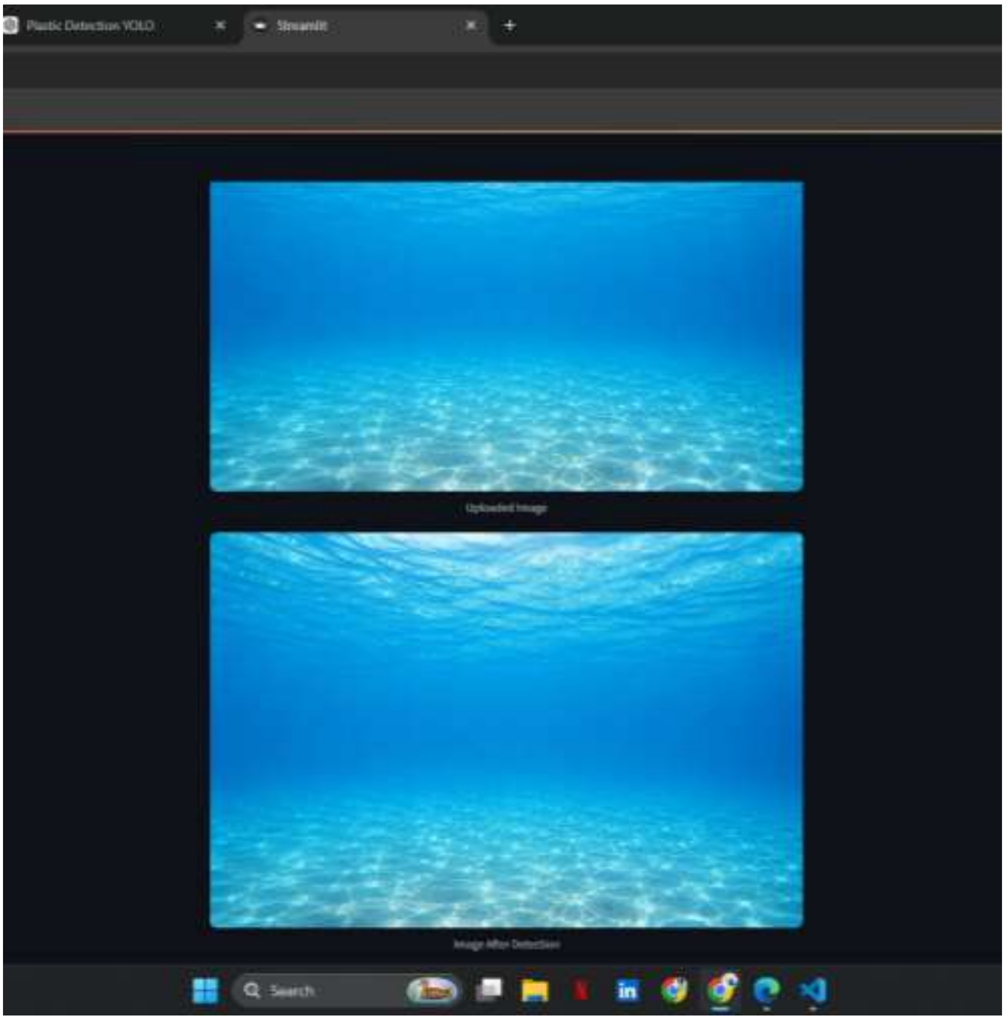


Figure 7.4: Prediction results for Test Image - 01

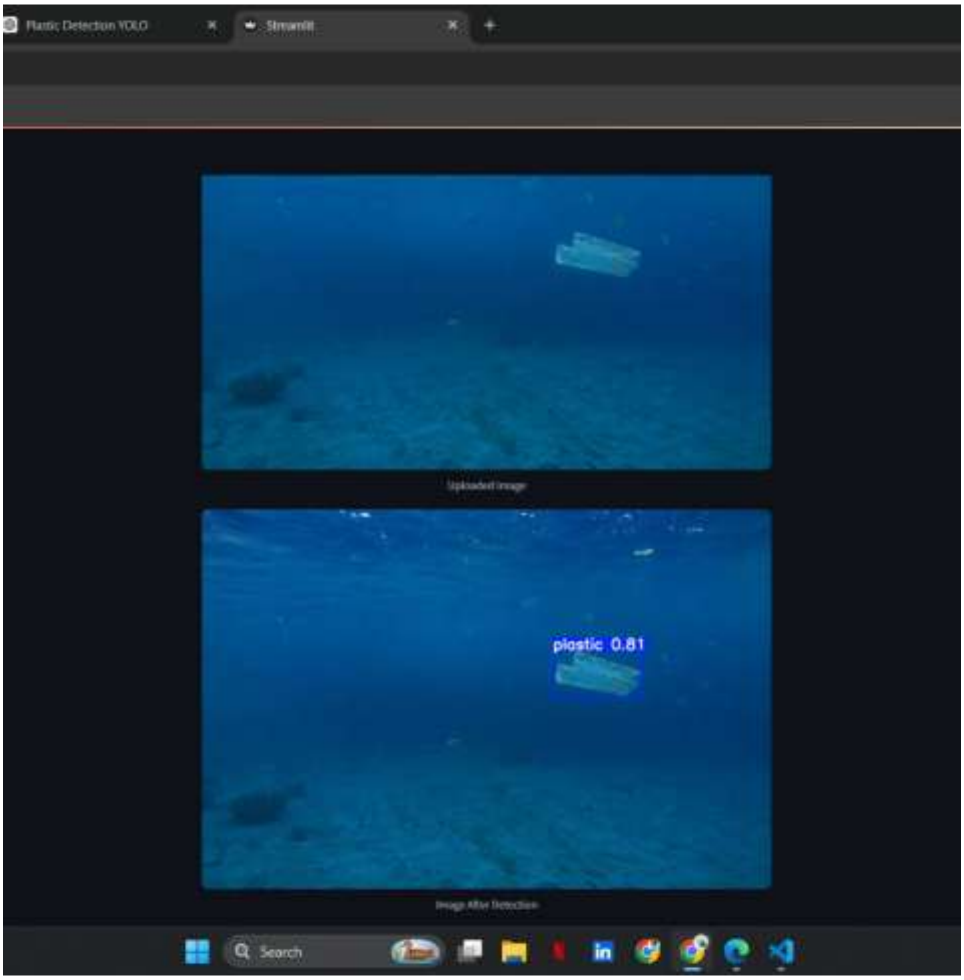


Figure 7.5: Prediction results for Test Image - 02

7.3 Performance Analysis

The performance of the Plastic Detection Using YOLO system was evaluated based on its detection capability, prediction speed, and usability. The YOLOv8 architecture provided efficient object detection while maintaining high processing speed suitable for practical applications.

The trained model successfully detected plastic waste in underwater images despite challenges such as varying object sizes, complex backgrounds, and underwater lighting conditions. The use of data augmentation techniques during training contributed to improved model generalization and robustness.

Key performance observations include:

- Fast image processing and prediction generation.
- Accurate localization of plastic waste using bounding boxes.
- Consistent performance across multiple test images.
- Efficient deployment through the Streamlit platform.
- Reduced manual effort in marine waste monitoring.
- Reliable detection results for environmental analysis.

The combination of YOLOv8 and the TrashCan dataset enabled the system to achieve effective object detection performance suitable for marine pollution monitoring applications.

7.4 Interpretation of Results

The obtained results indicate that the proposed Plastic Detection Using YOLO system is capable of effectively identifying plastic waste in underwater images. The model successfully learned the visual characteristics of plastic debris from the training dataset and applied this knowledge to detect similar objects in unseen images.

The generated bounding boxes accurately highlighted the locations of detected plastic waste, allowing users to easily identify areas affected by marine pollution. The confidence scores provided additional information regarding the reliability of each prediction.

. The results demonstrate that deep learning-based object detection can significantly improve the efficiency of marine waste monitoring compared to traditional manual inspection methods. The successful deployment of the model through a Streamlit interface further confirms the practical applicability of the proposed system for environmental monitoring, marine conservation, and research purposes.

The results obtained from the Plastic Detection Using YOLO system confirm the effectiveness of YOLOv8 for underwater plastic waste detection. The system successfully achieved its objectives by providing accurate detection, real-time prediction, and an accessible user interface. These outcomes demonstrate the potential of artificial intelligence and computer vision technologies in addressing marine plastic pollution and supporting environmental conservation efforts

CONCLUSION

The Plastic Detection Using YOLO project was successfully designed, developed, and implemented to provide an automated solution for detecting plastic waste in underwater environments. The system utilizes the YOLOv8 object detection model to identify and localize plastic debris present in underwater images with high efficiency. By employing deep learning and computer vision techniques, the proposed system reduces the dependency on manual inspection methods and improves the speed and accuracy of marine waste detection.

The model was trained using the TrashCan dataset and demonstrated reliable performance in detecting plastic waste under different underwater conditions. The integration of the trained model with a Streamlit-based web application enabled users to easily upload images and obtain detection results through a simple and interactive interface. The generated bounding boxes and confidence scores provided clear visualization of detected plastic objects.

The testing and evaluation results confirmed that the developed system successfully meets the project objectives. The proposed solution contributes to environmental monitoring efforts by providing an effective tool for identifying marine plastic pollution. The project demonstrates how Artificial Intelligence and Computer Vision technologies can be utilized to support marine conservation and environmental sustainability initiatives.

FUTURE ENHANCEMENTS

Although the Plastic Detection Using YOLO system successfully detects plastic waste in underwater images, several enhancements can be incorporated in future versions to improve its functionality, performance, and applicability.

- Extend the system to classify multiple categories of marine waste such as metal, glass, paper, and fishing nets.
- Implement real-time plastic detection using live underwater video streams.
- Integrate the system with underwater drones and autonomous robotic platforms for automated waste monitoring.
- Improve detection accuracy by training the model on larger and more diverse underwater datasets.
- Develop a cloud-based deployment platform for large-scale environmental monitoring applications.
- Incorporate advanced deep learning architectures to enhance detection performance under challenging underwater conditions.
- Enable automatic waste quantification and pollution assessment for environmental analysis and reporting.
- Develop a mobile application version of the system for improved accessibility and field usage.
- Integrate GPS and mapping technologies to track pollution hotspots and generate geographical waste distribution reports.
- Expand the system to support comprehensive marine ecosystem monitoring and conservation activities.

These enhancements can further improve the effectiveness of the system and enable its adoption in large-scale marine pollution monitoring and environmental protection initiatives.

REFERENCES

[1] J. Hong, M. Fulton, and J. Sattar, “TrashCan: A Semantically-Segmented Dataset towards Visual Detection of Marine Debris,” in IEEE International Conference on Robotics and Automation (ICRA), 2020. Available: https://arxiv.org/abs/2007.08097

[2] Z. Moorton, Z. Kurt, and W. L. Woo, “State of the Art Applications of Deep Learning within Tracking and Detecting Marine Debris: A Survey,” Marine Pollution Bulletin, vol. 199, 2024. Available: https://www.sciencedirect.com/science/article/pii/S0025326X24000234

[3] SeaClear Project Consortium, “A Dataset for Detection and Segmentation of Underwater Marine Debris in Shallow Waters,” 2024. Available: https://lcas.lincoln.ac.uk/wp/seaclear-dataset/

[4] J. Liu and Y. Zhou, “Marine Debris Detection Model Based on the Improved YOLOv5,” Journal of Marine Science and Engineering, vol. 11, no. 8, 2023. Available: https://www.mdpi.com/journal/jmse

[5] J. Zhang, J. Zhang, K. Zhou, Y. Zhang, H. Chen, and X. Yan, “An Improved YOLOv5-Based Underwater Object Detection Framework,” Applied Sciences, vol. 13, no. 5, 2023. Available: https://www.mdpi.com/journal/applsci

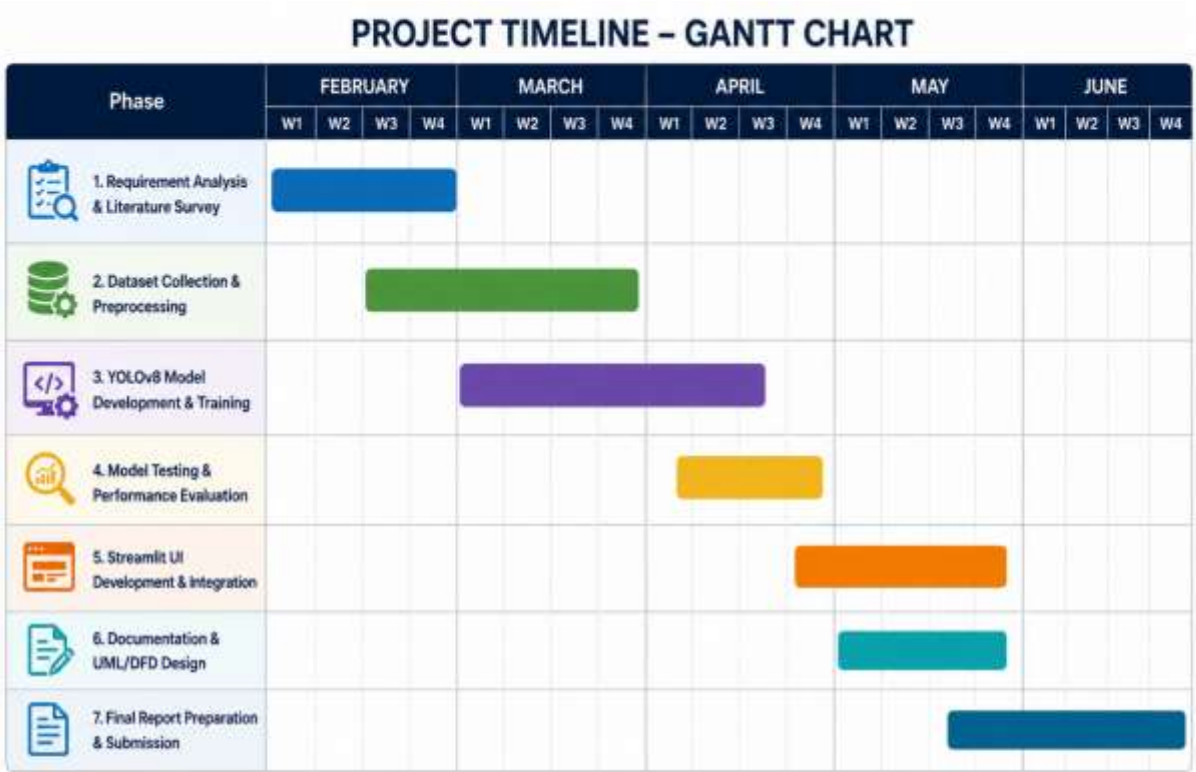
[6] J. Zhu, T. Hu, L. Zheng, N. Zhou, H. Ge, and Z. Hong, “YOLOv8-C2f-Faster-EMA: An Improved Underwater Trash Detection Model Based on YOLOv8,” Ocean Engineering, vol. 302, 2024. Available: https://www.sciencedirect.com/journal/ocean-engineering

[7] W. Jiang, L. Yang, and Y. Bu, “Research on the Identification and Classification of Marine Debris Based on Improved YOLOv8,” Journal of Marine Science and Engineering, vol. 12, no. 2, 2024. Available: https://www.mdpi.com/journal/jmse

APPENDIX

A. GANTT CHART

Project Progress Timeline (14 Weeks)



B. SAMPLE CODE

1. Dataset download using KaggleHub

```
import kagglehub
path = kagglehub.dataset_download(
    "mexwell/trashcan-1-0"
)
print("Dataset downloaded at:")
print(path)
```

2. COCO to YOLO annotations conversions

```
x_center = (x + w/2) / img_w
y_center = (y + h/2) / img_h
w = w / img_w
h = h / img_h
class_id = 0
line = f"{class_id} {x_center} {y_center} {w} {h}"
```

3. Dataset configuration file creation

```
data_yaml = """
path: /content/yolo_dataset
train: images/train
val: images/val
names:
  0: plastic
"""
with open(
    "/content/yolo_dataset/data.yaml",
    "w"
) as f:
    f.write(data_yaml)
```

4. **YOLOv8 model training**

```
model = YOLO("yolov8m.pt")
model.train(
    data="/content/yolo_dataset/data.yaml",
    epochs=20,
    imgsz=640,
    batch=16,
    patience=3,
    mosaic=1.0
)
```

5. **Platic waste detection using trained model**

```
results = model.predict(
    source=random_images,
    conf=0.15,
    save=True
)
```

6. **Streamlit based UI**

```
uploaded_file = st.file_uploader(
    "Upload Image",
    type=["jpg", "jpeg", "png"]
)
if uploaded_file is not None:
    image = Image.open(uploaded_file)
    results = model.predict(
        conf=0.25
    )
    plotted_image = results[0].plot()
    st.image(
        plotted_image,
        caption="Image After Detection",
        channels="BGR"
    )
```

C. PROJECT MAPPING

QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)			
Course Name:	MINI PROJECT	**AY**	2025-26
Course Code:	7PW663ML	**Semester**	VI
Name of the Guide:	Mr. MUJTHABA GULAM MUQEETH	**Class**	AI&ML - A
PROJECT TITLE **PLASTIC DETECTION USING YOLO**			

STUDENTS		
S. No	**HT No**	**Name of the Student**
1.	160723748001	Bharath Simha Reddy V
2.	160723748030	Raavi Indu
3.	160723748031	Korimi Akash

Course Outcome: After the successful completion of this course, the student will be able to:

CO No	Course Outcome	BTL	POs, PSOs Mapped
7PW663ML.1	Demonstrate the ability to synthesize and apply the knowledge and skills acquired in the academic program to the real-world problems.	**Apply**	PO1, PO2, PO3, PO6, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.2	Analyze and formulate the problem, and design solution of the selected project.	**Analyze**	PO1, PO2, PO3, PO4, PO5, PO6, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.3	Evaluate different solutions based on economic and technical feasibility.	**Evaluate**	PO3, PO4, PO5, PO6, PO7, PO8, PO11, PSO1, PSO2, PSO3
7PW663ML.4	Effectively plan, develop and deploy a project and confidently perform all aspects of project management.	**Create**	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2, PSO3
7PW663ML.5	Demonstrate effective written and oral technical communication skills.	**Create**	PO2, PO5, PO7, PO9, PO10, PO12, PSO1, PSO2, PSO3

CO-PO-PSO MAPPING

3 – High, 2 – Moderate, 1 – Slight

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
7PW663ML. 1	3	2	2	-	-	3	-	2	-	-	2	-	3	3	3
7PW663ML. 2	3	3	2	3	2	2	-	2	-	-	2	-	3	3	3
7PW663ML. 3	-	-	3	3	2	2	2	2	-	-	3	-	3	2	2

7PW663ML.4	-	-	3	3	3	2	2	3	3	3	3	2	3	3	3
7PW663ML.5	-	2	-	-	2	-	2	2	3	3	2	2	2	2	2

CO-PO-PSO MAPPING (WITH PERFORMANCE INDICATORS)

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS0 1	PSO 2	PSO 3
7PW663ML.1	1.1.2 1.2.1 1.3.1 1.4.1	2.1.2 2.1.3 2.2.3 2.3.1 2.4.1	3.1.1 3.1.3 3.1.4 3.1.5 3.2.1 3.2.3 3.3.1 3.3.2	-	-	6.1.1 6.2.1 6.3.1 6.3.2 6.4.1	-	8.2.1 8.2.2 8.2.3	-	-	11.1.2 11.2.1 11.2.2 11.3.1 11.3.2	-	**High**	**High**	**High**
7PW663ML.2	1.1.2 1.2.1 1.3.1 1.4.1	2.1.1 2.1.2 2.1.3 2.2.3 2.3.1 2.4.1 2.4.3 2.4.4	3.1.1 3.1.3 3.1.4 3.1.5 3.1.6 3.2.1 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.1 4.1.2 4.1.3 4.2.1 4.3.1 4.3.2 4.3.3 4.3.4	-	6.1.1 6.3.1 6.3.2 6.4.2	-	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	-	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**High**	**High**
7PW663ML.3	-	2.2.2 2.2.3 2.2.4 2.3.2 2.4.2 2.4.3 2.4.4	3.1.5 3.2.1 3.2.2 3.2.3 3.3.1 3.3.2 3.4.1 3.4.2	4.1.2 4.2.1 4.2.2 4.3.1 4.3.2 4.3.3 4.3.4	5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	-	10.1.1 10.1.2 10.2.1	11.1.2 11.2.1 11.2.2 11.3.2	-	**High**	**Mod**	**Mod**
7PW663ML.4	-	2.3.1 2.3.2 2.4.1 2.4.2 2.4.3 2.4.4	3.1.6 3.2.1 3.2.2 3.3.2 3.4.1 3.4.2	4.1.2 4.1.3 4.2.1 4.2.2 4.3.1 4.3.3	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	6.1.1 6.2.1 6.3.1 6.3.2 6.4.2	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**High**	**High**	**High**
7PW663ML.5	-	2.2.1 2.2.2 2.2.3 2.2.4	-	-	5.1.1 5.1.2 5.2.1 5.2.2 5.3.1 5.3.2	-	7.1.1 7.1.2 7.2.1 7.2.2	8.1.1 8.1.2 8.2.1 8.2.2 8.2.3 8.3.1	9.1.1 9.1.2 9.1.3 9.2.1 9.2.2 9.3.1 9.3.2	10.1.1 10.1.2 10.2.1 10.3.1 10.3.2	11.1.1 11.1.2 11.2.1 11.2.2 11.3.2	12.1.1 12.1.2 12.2.1 12.2.2 12.3.1 12.3.2	**Mod**	**Mod**	**Mod**

MINI PROJECT SPECIFIC LEARNING OUTCOMES

LO#	Outcome Description
1	Understand and apply machine learning and computer vision concepts for environmental monitoring applications.
2	Collect, preprocess, and prepare underwater image datasets for object detection model training.
3	Develop and train a YOLOv8-based deep learning model for marine plastic waste detection.
4	Design and implement a user-friendly web application using Streamlit for real-time image-based plastic detection.

5	Evaluate system performance and analyze detection results to support marine conservation and pollution monitoring efforts.

PO -WK MAPPING:

PO#	PO Description	Mapped WKs	PO#	PO Description	Mapped WKs
PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Environment and sustainability	WK5, WK7
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Ethics	WK7, WK9
PO3	Design / Development of Solutions	WK1, WK2, WK3, WK4, WK5, WK6, WK7	PO9	Individual & Team Work	WK9
PO4	Conduct Investigations of complex problems	WK2, WK3, WK4, WK8	PO10	Communication	WK7, WK9
PO5	Engineering Tool usage	WK2, WK6	PO11	Project Management & Finance	WK5, WK6, WK7
PO6	The Engineer and society	WK1, WK5, WK7	PO12	Life-long Learning	WK8

SDG MAPPING:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
1 NO POVERTY		7 AFFORDABLE AND CLEAN ENERGY		13 CLIMATE ACTION	13.3.1
2 ZERO HUNGER		8 DECENT WORK AND ECONOMIC GROWTH		14 LIFE BELOW WATER	14.1.1
3 GOOD HEALTH AND WELL-BEING		9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	9.5.1	15 LIFE ON LAND	15.1.1
4 QUALITY EDUCATION		10 REDUCED INEQUALITIES		16 PEACE, JUSTICE AND STRONG INSTITUTIONS	
5 GENDER EQUALITY		11 SUSTAINABLE CITIES AND COMMUNITIES		17 PARTNERSHIPS FOR THE GOALS	
6 CLEAN WATER AND SANITATION		12 RESPONSIBLE CONSUMPTION AND PRODUCTION	12.5.1		

QUALITY ANALYSIS:

MINI PROJECT COLLABORATION

(Tick ✓ Whichever is applicable and present the details below):

Industry	Research Lab	Professional Society	Community	Alumni	Regulating agency	Govt.	Other
			✓				

MINI PROJECT ACHIEVEMENTS:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes

S. No	Description	Details
1.	Prototype development	Developed a working prototype for underwater plastic waste detection using YOLOv8.
2.	Technology based solution developed	Implemented an AI-powered computer vision system for marine plastic detection.
3.	Publication	Nil
4.	Start-up incubation	Nil
5.	Competition prizes	Nil
6.	Community / Industry / HEI service	Supports environmental monitoring and marine conservation initiatives.
7.	SDG Indicators	SDG 14, SDG 13, SDG 12, SDG 15, SDG 9.
8.	Grants / Financial Aid	Nil
9.	Multi-disciplinary	Combines Artificial Intelligence, Machine Learning, Computer Vision, and Environmental Science.
10.	Internship / Employment	Provides practical experience in AI-based object detection and application development.
11.	Self-employment skill	Enhances skills in Machine Learning, Deep Learning, Python, YOLOv8, and Streamlit deployment.
12.	Use of real-time data	Utilizes real underwater image datasets (TrashCan Dataset) for model training and evaluation.

Signature of the Guide **Signature of the Project Coordinator**

(QR CODE with all project docs in Gdrive)